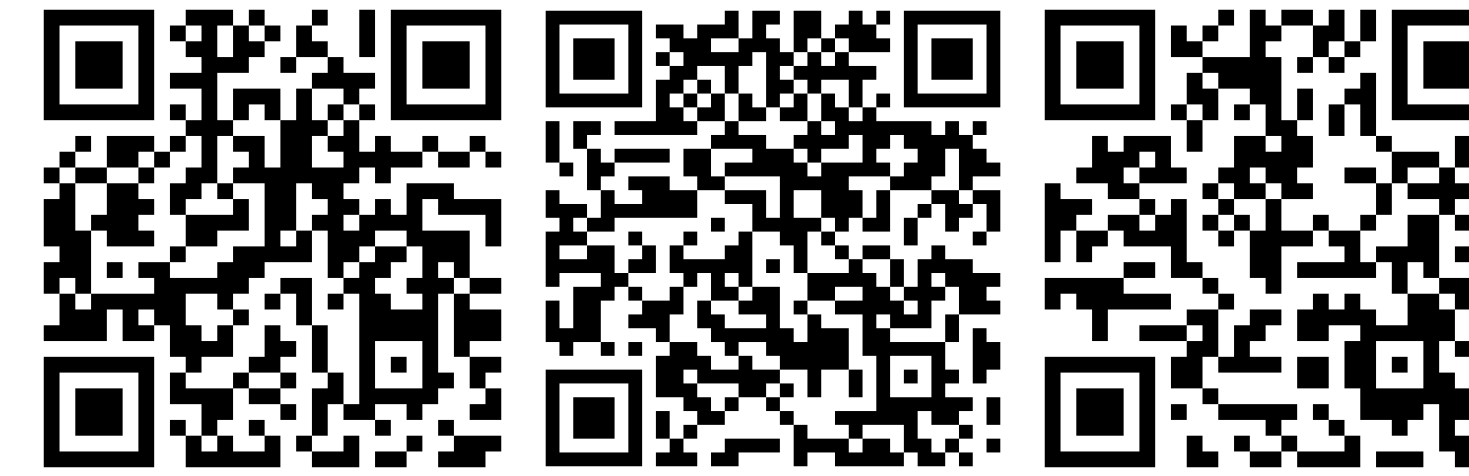
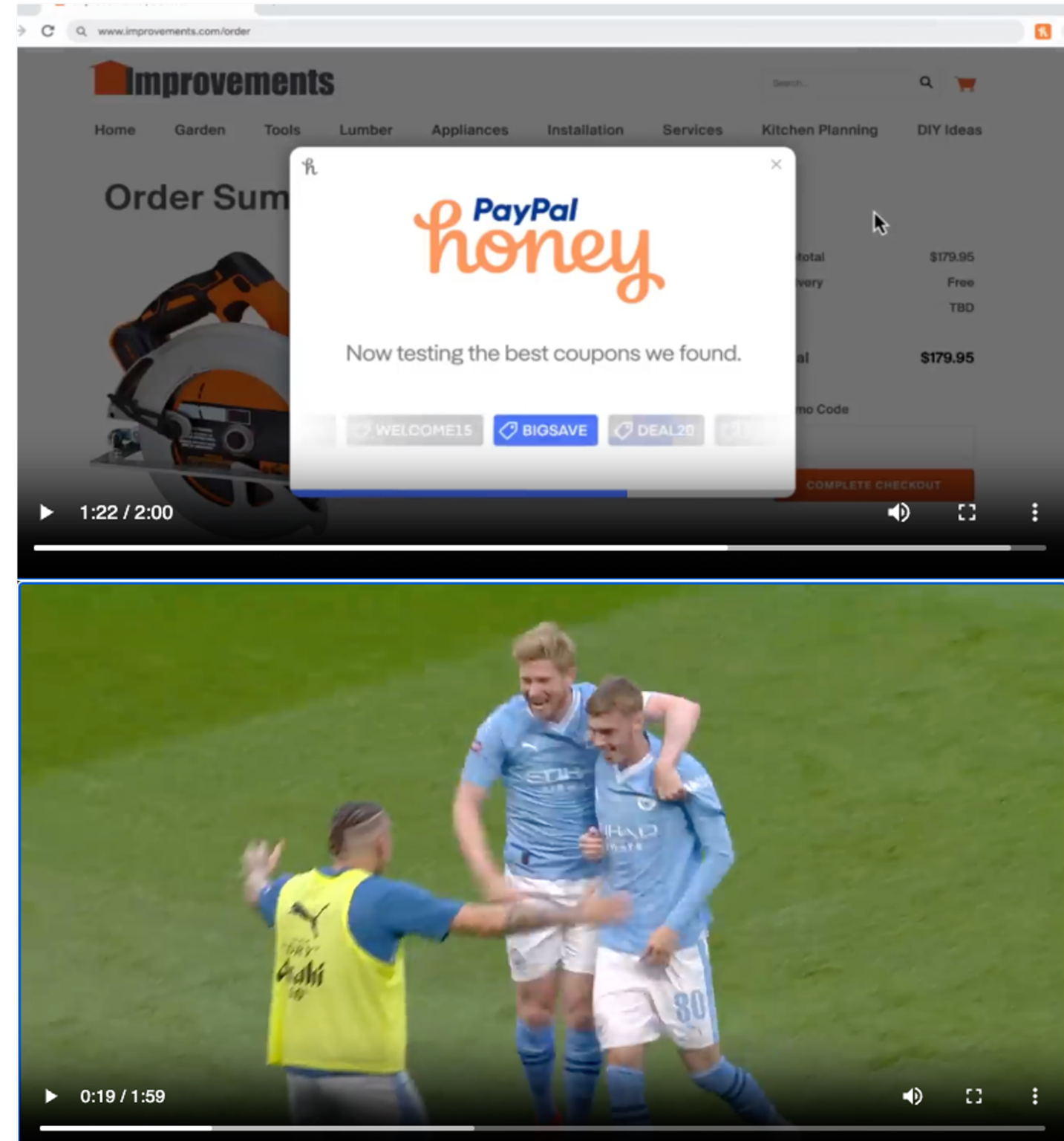




Motivation

What is Modality Dominance?

VLMs rely on linguistic patterns than true visual understanding.



Question: Which browser plugin does the blogger recommend?

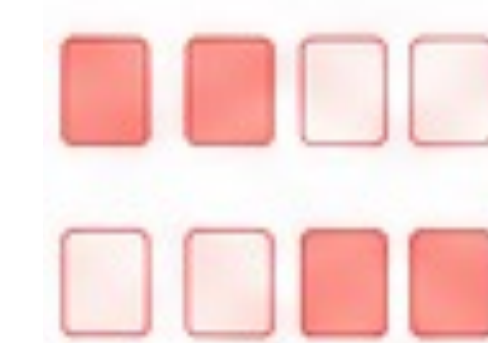
Baseline: He recommends using the website plugin.
Ours: He recommends using the honey.com.
GT: Paypal Honey.

Question: Summarize the video.

Baseline: Of course, I see there is a video.
Ours: The video captures a soccer match in progress. It describes the passionate hug between two.
GT: This video is a key moments replay of a football game. The team in red jerseys scored two goals, and the team in blue jerseys scored one goal.

Why Modality Dominance?

Not every modality contains sufficient information.



Sparse Vision Tokens

"What is the woman currently making?"



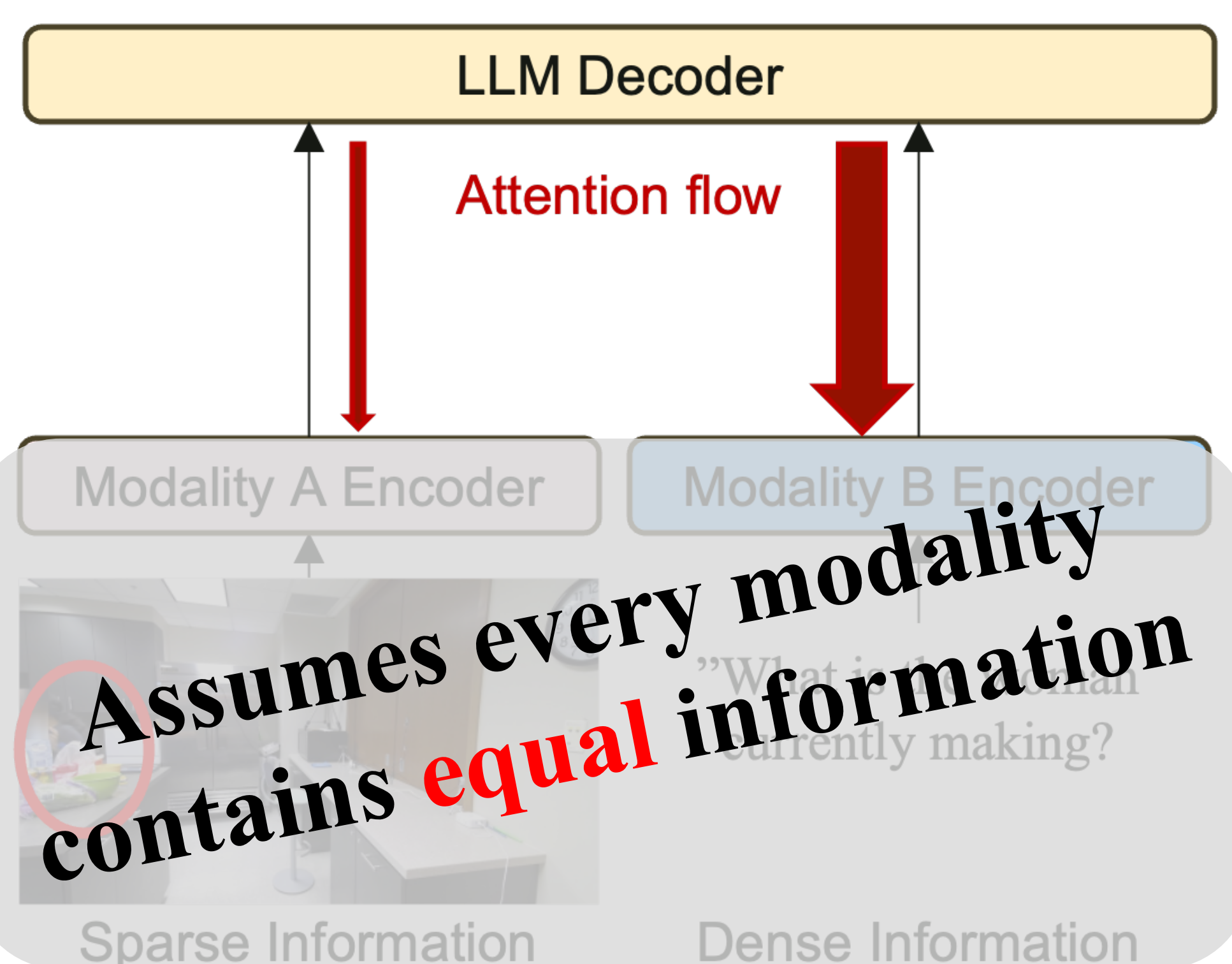
Dense Language Tokens



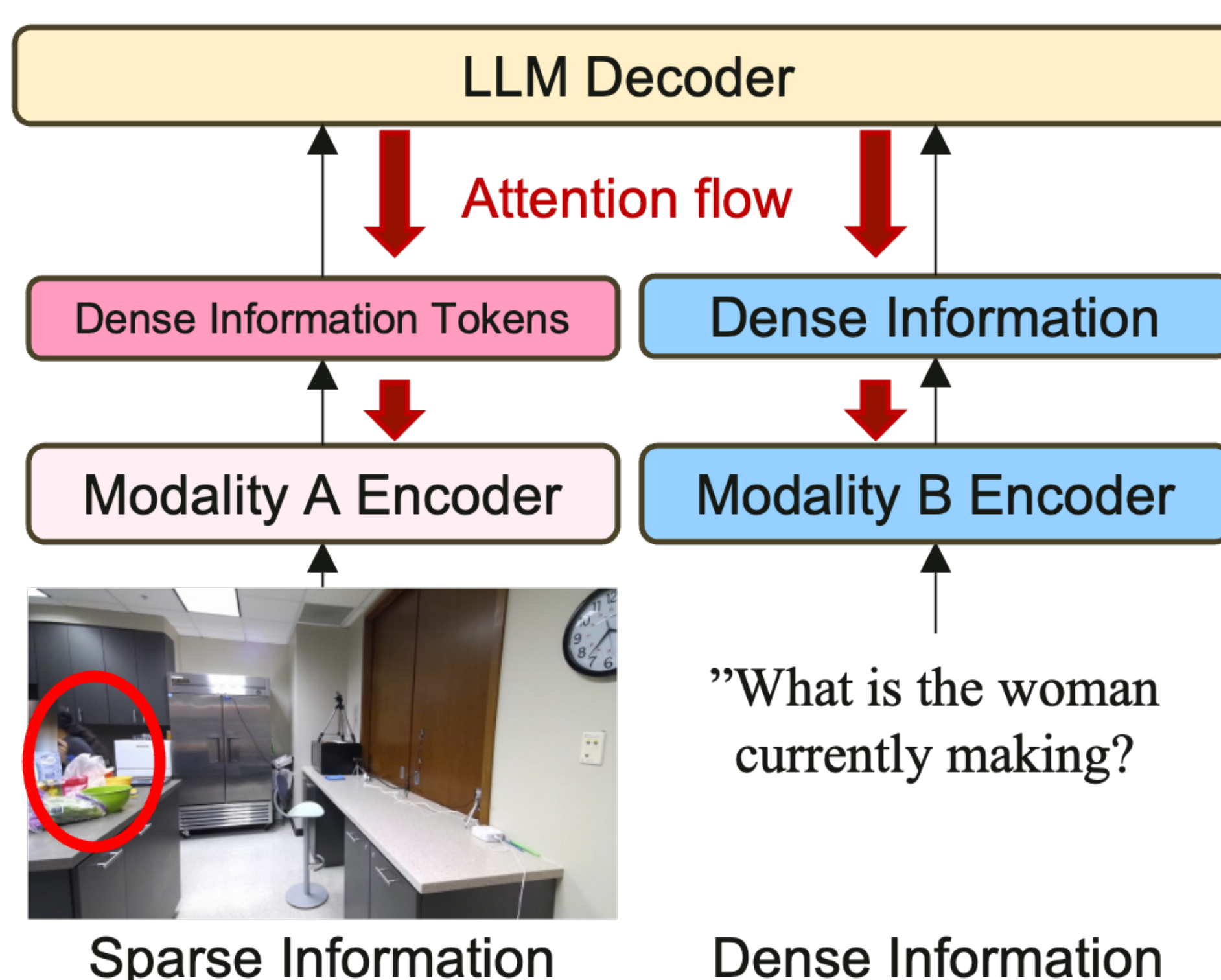
Disparity

Contribution

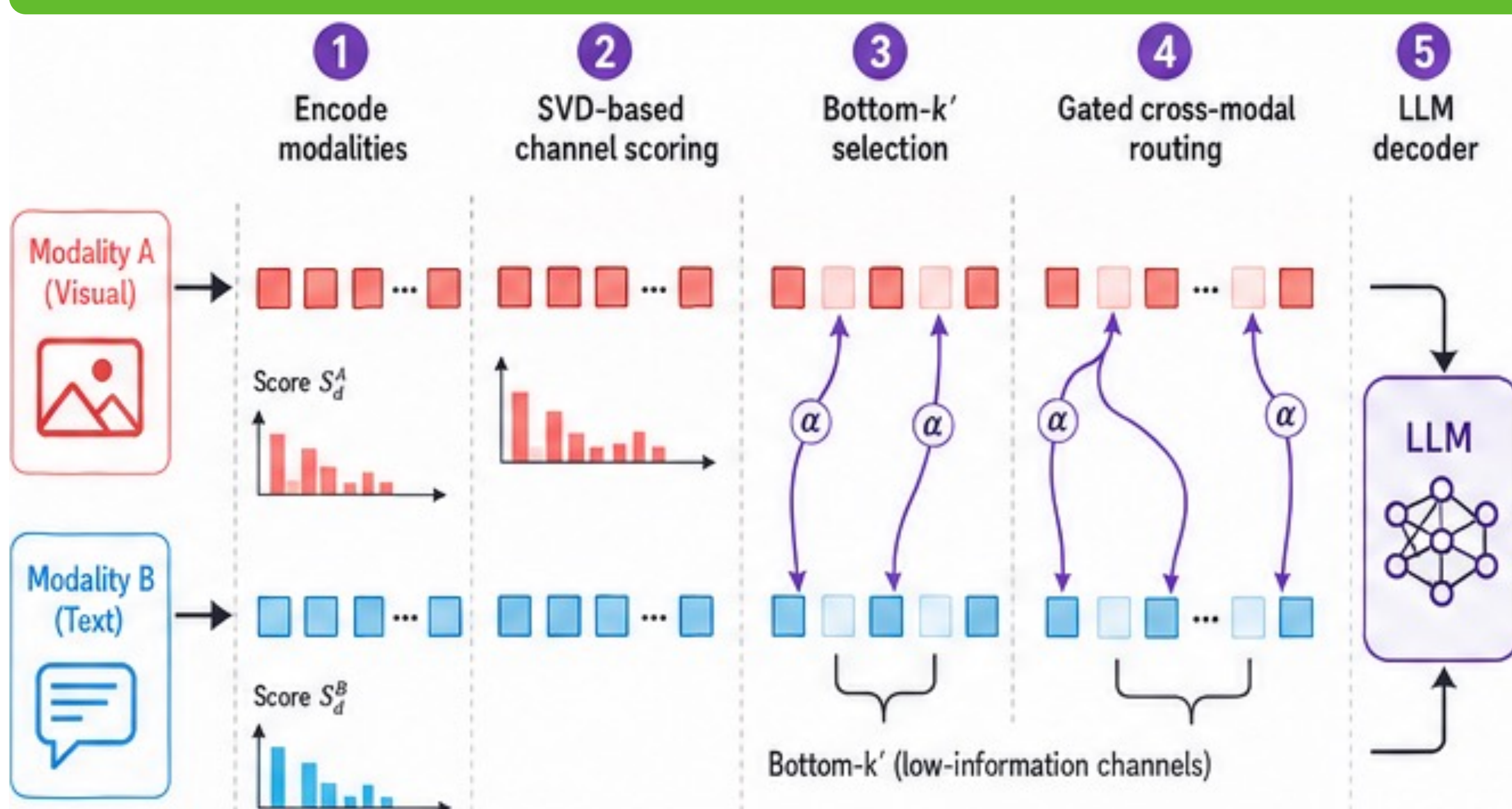
Previous Method



Our Method



Architecture



2 Channel Informativeness

$$S^d = \sum_i \sigma_i^2(v_{i,d})^2$$

3 Less-informative channels

$$D_{low} = Bottom - k'(S)$$

4 Gated cross-modal routing

$$routed = \alpha \cdot complement + (1 - \alpha) \cdot original$$

Results

higher accuracy



lower MDI + AEI



REPRESENTATIVE ACCURACY



VISUAL CORRUPTION ROBUSTNESS

Lower unchanged-prediction rate = stronger visual reliance

